Uroš Savurdić

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Serbian and Croatian/EU citizen — no work permit needed in Serbia, the EU/EEA or Switzerland; open to relocation and sponsorship elsewhere.

M.Sc. EECS student and former Applied Science Intern at Microsoft. I build and evaluate LLM systems end to end — retrieval and fine-tuning pipelines, evaluation harnesses, and the measurement work that shows whether any of it actually helps.

EXPERIENCE

Microsoft – Applied Science Intern · LLMs · evaluation · knowledge graphs · graph search · multi-hop reasoning Mar 2026 – Jul 2026

- Built a selective two-view framework for the difficulty of generated multi-hop questions — a structural ranker before generation, an independent surface grader after. Where they agree (65.5% of 160 expert-labelled questions from 45 documents) they matched expert labels 81.2% of the time, against 64.6% for hop count at the same coverage: a paired +16.6 points, 95% CI 5.8–28.4.
- Built an end-to-end pipeline turning long enterprise documents into evidence-grounded multi-hop QA benchmarks: schema-free two-pass knowledge-graph extraction with a grounding filter against fabricated entities, MCTS and guided-walk graph search, and LLM generation with evidence traceability.
- Ran a controlled search experiment (three paired seeds, 2,000 evaluations per document and policy) showing that optimizing the learned difficulty proxy with MCTS and guided walks raised the proxy far more than independently graded difficulty.
- Showed through controlled ablations that removing grounded extraction raised predicted Hard+ difficulty from 35.6% to 87.8% while lowering answerability, while swapping in a smaller extraction model had little effect — construction choices mattered more than model size.
- Hardened document ingestion across 218 documents, recovering 4,317 footnotes the parser had silently dropped by reading the DOCX XML directly. Built cross-chunk entity canonicalization reaching F1 0.97 on 20 DocRED documents in development testing.

IPSI Belgrade – Research Collaborator · with Prof. Veljko Milutinović Feb 2026 – present

- Working with Prof. Veljko Milutinović on a comparative survey of emerging computing paradigms for AI workloads.
- Running the literature review and contributing technical review for the survey and an accompanying book.

RESEARCH

Internal Representations of Safety Across Post-Training · Independent research · TELFOR 2026 paper · PyTorch, LoRA, TRL Aug – Oct 2026

- Compared safety SFT and DPO on Qwen2.5-1.5B from matched instruction-tuned starts and the same 4,000 PKU-SafeRLHF pairs: two starts, three training runs per arm.
- Found that SFT preserves the model's internal harmful–benign contrast while DPO redirects it: from layer 20 of 28 on, most of the contrast lies orthogonal to the original, in every run, although SFT changed the weights 6.7× as much.
- Evaluated behaviour with LLM judges validated on the benchmark (κ 0.95) and a blind human check: DPO removed nearly all harmful answers but refused 24–53% of benign prompts that only sound alarming.

Attention Mechanisms Across Computational Paradigms · Benchmarking and profiling · PyTorch, JAX, NumPy Apr – May 2026

- Benchmarked attention on 12 execution substrates and 17 implementations across four computing paradigms, decomposed into five stages and measured for FLOPs, memory traffic and arithmetic intensity. Measured on CPU, an NVIDIA T4 and a TPU, and simulated or modelled analytically for emerging hardware, with each model checked against published device data.
- Traced a 20x FP32 slowdown to NumPy 2 silently promoting float32 inputs to float64, and verified it with clean FP32/FP64 microbenchmarks (4,002 GFLOP/s for attention on a T4).
- Built the harness as an extensible package: an abstract paradigm interface, a registry decorator, dataclass metrics and a structured experiment logger.

Presented at the 5th Serbian International Conference on Applied Artificial Intelligence (SICAAI), Kragujevac, May 2026.

PROJECTS

Code Search Engine with Contrastive Learning · PyTorch Lightning, FAISS, Sentence-Transformers Nov 2025

- Fine-tuned a bi-encoder (all-MiniLM-L6-v2) with InfoNCE contrastive loss on 20k CoSQA query-code pairs, improving MRR@10 from 0.786 to 0.911 (+16.0%), with NDCG@10 +12.1% and Recall@10 +2.8% over the pretrained baseline.
- Enabled semantic search across 550+ indexed code snippets with FAISS and cosine similarity.
- Set up a reproducible training and evaluation pipeline with PyTorch Lightning, AdamW with warmup, Weights & Biases tracking, and a torchmetrics evaluation suite.

Handwritten Text Recognition System · PyTorch, Transformers, Docker, AWS Aug – Oct 2025

- Built and deployed an end-to-end recognition pipeline: CNN-LSTM and ResNet-Transformer models trained on a modified IAM dataset, with experiment tracking in Weights & Biases, hyperparameter tuning and CI/CD through CircleCI.
- Shipped it as a containerized FastAPI service architected for AWS Lambda.

Applied ML and Data Science Projects · Python, XGBoost, SciPy, LLM APIs Mar – Oct 2025

- Predicted user engagement to an MAE of 5.06 by engineering 30+ temporal and behavioral features from 90k sessions.
- Evaluated A/B tests using both frequentist and Bayesian methods.
- Automated natural-language queries with intent classification, context extraction, and temporal and geolocation logic.

Optimization Methods · NumPy, from scratch Nov – Dec 2025

- Implemented forward and backward passes by hand in NumPy - manual softmax and ReLU gradients, cross-entropy with log stabilisation - rather than calling an autograd framework.
- Compared gradient descent against a genetic algorithm for training the same network, plotted against wall-clock time so the comparison reflects real cost.
- Implemented 2-opt local search and exhaustive search for a travelling-salesman drilling problem — the heuristic matches the proven optimum on every instance small enough to enumerate, about 70,000x faster at 12 points — plus simulated annealing, a genetic algorithm and differential evolution on constrained problems.

PUBLICATIONS

- U. Savurdić, "Matched Starts, Different Recipes: Chosen-Response SFT and Direct DPO Change a Pre-Existing Benchmark-Defined Activation Contrast." Accepted at TELFOR 2026, the Telecommunications Forum, Belgrade.
- Analysis of Attention Mechanisms in the Context of Computational Paradigms — presented at the 5th Serbian International Conference on Applied Artificial Intelligence (SICAAI), Kragujevac, May 2026.

TECHNICAL SKILLS

Languages	Python, SQL, C/C++
ML	PyTorch, Transformers, Hugging Face, LoRA/PEFT, TRL, sentence-transformers, JAX
LLM systems	fine-tuning, DPO, RAG and retrieval, FAISS, evaluation harnesses, LLM-as-judge
Infrastructure	Docker, AWS, FastAPI, Git, CI/CD, Weights & Biases
Spoken	Serbian/Croatian — native · Macedonian — native · English — fluent · Russian and Bulgarian — understood · German — basic

EDUCATION

University of Belgrade — School of Electrical Engineering (ETF) expected Sep 2027
M.Sc. Electrical Engineering & Computer Science · GPA 10/10 · evaluation, system security, optimization

University of Belgrade — School of Electrical Engineering (ETF) graduated Sep 2024
B.Sc. Electrical Engineering & Computer Science · signal processing, information theory

Additional coursework: Full Stack Deep Learning (UC Berkeley); Statistical Learning and CME295 Transformers & LLMs (Stanford).

VOLUNTEERING

EESTEC (Electrical Engineering Students' European Association) Feb 2019 – present

- Delivered 100+ hours of training to 400+ students across Europe and mentored around 30 members into organizational roles; certified Soft Skills Trainer (2023).
- Led a 30-member international team through the COVID-19 transition. Named EESTECer of the Year 2021 across a 4,000+ member network.

AWARDS

EESTECer of the Year — Board of the Association, one recipient per year across a 4,000+ member network (2021) · 3rd place, National Mathematics Competition — Mathematical Society of Serbia (2017) · Best Student Start-Up Award — Science-Technology Park Belgrade (2022)